

MASSACHUSETTS INDUSTRIAL PROPERTIES PRICE PREDICTION AND MARKET ANALYSIS



MIS587 - Business Applications in Machine Learning
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Presented to: **LORNELL REAL ESTATE**

PROBLEM STATEMENT & PROJECT GOALS



THE CHALLENGE:

- Industrial real estate market relies heavily on historical comparables and qualitative assessments
- Subtle market dynamics and hidden opportunities are often overlooked



OUR SOLUTION:

- Leverage machine learning to provide quantitative, data-driven insights for fair asset valuation

PROJECT ORGANIZATION



STAKEHOLDERS & CUSTOMERS:

- Todd Lornell - Principal/ Founder Lornell Real Estate
- Commercial Real Estate Brokers
- Real Estate Investment Analysts
- Acquisitions Managers



DAY-TO-DAY PROJECT TEAM:

- Shrey Sharma - Project Manager
- Martin Thulani Milanzi - Risk Analyst
- Alex Siracusa - Lead Data Analyst
- Faisal Yaseen - Subject Matter Expert

PROJECT OBJECTIVES



1. ACCURATE PRICE PREDICTION

Develop models to accurately predict industrial property prices, especially price per square foot



2. MARKET TIMING FORECAST

Forecast Time-on-Market to inform bidding strategies and negotiation tactics



3. VALUE ATTRIBUTION ANALYSIS

Quantify specific features that drive value using regression and conjoint analysis



4. OPPORTUNITY IDENTIFICATION

Identify promising off-market properties through predictive framework analysis

METHODOLOGY: DATAROBOT PLATFORM



ENTERPRISE AI PLATFORM FEATURES:

- Comprehensive automated machine learning workflow for rapid model development
- Competitive process across various algorithms including gradient-boosted trees and deep learning
- Automated training and evaluation of diverse predictive models
- Ensures selection of most accurate and robust model
- Enhances project efficiency and minimizes deployment of suboptimal models

DATA ACQUISITION & EXPLORATION



DATA SOURCE: COSTAR

- User login credentials required
- Export format: .xlsx files (limited to 500 records per transaction)



DATASET CHARACTERISTICS:

- 15,467 commercial property records
- 87 features after data cleaning (from 100+ original columns)



FEATURE MIX:

- 42 numerical variables | 22 categorical variables | 15 location variables | 8 temporal variables

DATA EXPLORATION: KEY AREAS



TARGET VARIABLE ANALYSIS

- Sale Price Distribution, Price per SF, Days on Market



MARKET TIMING FACTORS

- Pricing Strategy Analysis, Property Marketability, Market Conditions



PRICE DRIVERS EXPLORATION

- **Location-Based:** Market Name, Submarket, City, State
- **Property Characteristics:** RBA, Building Age, Class, Star Rating
- **Industrial Features:** Ceiling Height, Loading Docks, Cranes, Drive-ins



DATA QUALITY ASSESSMENT

- Missing data patterns, Feature correlations, Outlier detection

FEATURE ENGINEERING: OVERVIEW



LEVERAGING DATAROBOT CAPABILITIES:

- Automated feature engineering to generate new predictive features
- Discovery of complex interactions, transformations, and encodings
- Example interactions: Year Built $\times$ Submarket, temporal data transformations
- Uncover non-linear relationships to improve model accuracy
- Systematic approach to feature selection using L1 regularization, Random Forest importance, and SHAP values

FEATURE ENGINEERING: KEY CATEGORIES



PROPERTY AGE & CONDITION

- Building Age, Years Since Renovation, Effective Age



FINANCIAL & RETURN METRICS

- Annual Revenue, Implied Cap Rate, NOI Estimate, Price Appreciation Rate



LOCATION & MARKET CONTEXT

- Market Tier, Submarket Price Rank, Geographic Clusters, Competitive Position



INDUSTRIAL FUNCTIONALITY

- Loading Docks per 10k SF, Drive-ins per 10k SF, High Ceiling Flag, Rail Access



OPPORTUNITY IDENTIFICATION

- Value-Add Flags, Distress Indicators, Renovation Opportunities, Rent Upside

RISK ANALYSIS: DATA QUALITY



CHALLENGES

- Imperfect real estate data (incomplete records, inaccuracies, inconsistencies)
- Missing key features and geographic bias in dataset
- Scarcity of off-market transaction data



MITIGATION STRATEGIES

- Rigorous data-cleaning and validation pipeline
- Statistical imputation methods with SME consultation
- Thorough exploratory data analysis to identify biases
- Explicit definition of model operational boundaries

RISK ANALYSIS: TECHNICAL & MODELING



CHALLENGES:

- Model drift due to dynamic market conditions
- Interpretability of "black box" machine learning models



MITIGATION STRATEGIES:

- Deploy solution on Microsoft Azure with continuous monitoring
- Automated alerts for model drift and algorithmic retraining triggers
- Integrate SHAP frameworks for feature-based prediction explanations
- Build trust through transparent, interpretable model outputs

RISK ANALYSIS: PROJECT & STAKEHOLDER MANAGEMENT



CHALLENGES:

- Delayed decision-making from stakeholders and SMEs
- Scope creep and low user adoption



MITIGATION STRATEGIES:

- Clear governance and communication plan with regularly scheduled stakeholder meetings
- Defined escalation path and formal change request process
- User-centric design methodology with end-user involvement
- Continuous feedback through interviews, prototype testing, and iteration

THANK YOU !

